Ariel University

Department of Computer Science

Estimating the Intrinsic Dimension of High-Dimensional Data

Course: Multivariate Probability

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Instructor: Dr. Elad Eigner-Horev

Submitted by:

Maya Rom

ID: 207485251

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1.Introduction

Intrinsic dimension refers to the minimal number of variables required to accurately represent the underlying structure of a dataset. Although many datasets are represented in high-dimensional ambient spaces, their true generative structure is often governed by a significantly smaller number of degrees of freedom. Estimating this intrinsic dimensionality is therefore an important step in understanding the geometric and statistical properties of complex data.

This report presents an intrinsic-dimension analysis conducted on four datasets: a two-dimensional plane embedded in a ten-dimensional space, the Swiss roll manifold, high-dimensional Gaussian noise, and the MNIST handwritten digit dataset. These datasets were selected to represent different types of data structure, including linear low-dimensional manifolds, nonlinear manifolds, unstructured high-dimensional data, and real-world image data.

To estimate intrinsic dimensionality, three complementary approaches were employed. Principal Component Analysis (PCA) was used to examine the number of linear components required to explain a large portion of the variance in the data. The correlation dimension was used to characterize local geometric scaling properties of the dataset. In addition, a k-nearest neighbors (kNN) based estimator was applied in order to obtain a neighborhood-based estimate of intrinsic dimensionality.

By combining these methods, the analysis aims to compare linear variance-based structure with nonlinear geometric properties, providing a more comprehensive understanding of dataset complexity.

1.1.Project Goal

The goal of this project is to analyze high-dimensional datasets and estimate their intrinsic dimensionality using several complementary statistical and geometric methods. In particular, the project applies PCA-based dimensionality estimation, correlation-dimension analysis, and a k-nearest neighbors maximum-likelihood estimator.

The analysis compares synthetic datasets with known or interpretable geometric structure to a real-world dataset in order to evaluate how different intrinsic-dimension estimators behave across varying data regimes. Through this comparison, the project aims to better understand how high-dimensional datasets differ in their geometric structure, effective complexity, and latent dimensionality.

1.2.Methods Overview

The intrinsic-dimension analysis in this project combines three complementary estimation approaches:

- Principal Component Analysis (PCA) was evaluated using cumulative explained variance curves and scree plots in order to determine how many linear components are required to represent most of the variance in the data.
- Correlation dimension was estimated from the slope of the linear scaling region in log-log correlation-integral plots, providing a geometric estimate of intrinsic dimensionality based on local distance scaling.
- k-Nearest Neighbors (kNN) estimation was performed using the Levina-Bickel maximum-likelihood estimator, which estimates intrinsic dimension from the distribution of nearest-neighbor distances.

Together, these methods provide complementary perspectives on dataset structure, allowing the analysis to compare linear variance-based representations with intrinsic geometric complexity.

2.Dataset Selection

2.1.Real-World Dataset

2.1.1.MNIST

The MNIST dataset was selected as the real-world benchmark for this study due to its well-established role in high-dimensional data analysis and machine learning research. In MNIST, each handwritten digit image is represented as a 28×28 grayscale image that is flattened into a vector of 784 features. Although the ambient dimensionality of the dataset is therefore relatively high, the underlying generative process of handwritten digits is expected to be governed by a much smaller number of latent factors, such as stroke thickness, writing style, digit shape, and spatial orientation.

This property makes MNIST a particularly suitable dataset for intrinsic-dimension analysis. The dataset is widely used, well-documented, and reliable, which allows results to be interpreted within a well-understood context. Moreover, MNIST combines high-dimensional representation with meaningful underlying structure, making it an effective real-world benchmark for evaluating intrinsic-dimension estimation methods.

2.1.2.Data Source

The MNIST dataset used in this study was obtained from the OpenML repository:

MNIST dataset (OpenML): https://www.openml.org/search?type=data&sort=runs&id=554&status=active

2.2.Synthetic Datasets

In addition to the real-world dataset, three synthetic datasets were included in the analysis:

- High-dimensional Gaussian noise
- A two-dimensional plane embedded in a ten-dimensional space
- The Swiss Roll manifold

These datasets were selected to provide controlled experimental scenarios with known or interpretable geometric structures.

2.2.1.Motivation for Synthetic Datasets

Synthetic datasets play an important role in evaluating intrinsic-dimension estimators because they allow the behavior of the estimators to be analyzed under controlled conditions. In such datasets, the expected intrinsic dimensionality is either known or theoretically well understood.

High-dimensional Gaussian noise was selected as a representative example of unstructured data in a high-dimensional space. Because the samples are generated from a multivariate Gaussian distribution without an underlying low-dimensional manifold, the estimated intrinsic dimensionality is expected to remain relatively high and close to the ambient dimension.

A two-dimensional plane embedded in a ten-dimensional space was chosen as a simple linear low-dimensional dataset with a known intrinsic dimensionality. Since the data are generated from only two degrees of freedom, this dataset provides an effective validation case for assessing whether the estimation methods can correctly recover a simple linear manifold structure.

The Swiss Roll dataset was selected as a canonical example of a nonlinear manifold. Although the data are embedded in a higher-dimensional space, the intrinsic structure of the dataset remains two-dimensional. This dataset is particularly useful for illustrating the limitations of linear dimensionality-reduction techniques such as PCA and for demonstrating the advantages of geometric intrinsic-dimension estimators.

2.2.2.Dataset Generation

The synthetic datasets were generated programmatically within the Python project environment.

- The Swiss Roll dataset was generated using sklearn.datasets.make_swiss_roll.
- High-dimensional Gaussian noise was generated using NumPy.
- The two-dimensional plane embedded in ten dimensions was constructed manually using a random linear embedding.

2.2.3.Final Dataset Configuration

The final experimental setup therefore includes one real-world dataset and three synthetic datasets. The real-world dataset is MNIST, while the synthetic datasets consist of high-dimensional Gaussian noise, a two-dimensional plane embedded in ten dimensions, and the Swiss Roll manifold.

Together, these datasets provide a balanced and informative framework for evaluating intrinsic-dimension estimation methods across several distinct data regimes: unstructured high-dimensional data, linear low-dimensional manifolds, nonlinear manifolds, and structured real-world datasets. This combination enables a comprehensive comparison of how different estimation techniques behave under varying geometric conditions.

3.Methods

3.1.k-Nearest Neighbors Based Intrinsic Dimension Estimation

In order to complement the PCA-based analysis and the correlation-dimension approach, an additional intrinsic-dimension estimator based on k-nearest neighbors (kNN) was employed. While PCA primarily captures global linear variance structure and the correlation dimension measures geometric scaling properties of the data distribution, kNN-based estimators provide a local geometric perspective on intrinsic dimensionality by analyzing the distribution of distances between neighboring data points.

In this project, intrinsic dimensionality was estimated using the Levina–Bickel maximum likelihood estimator (MLE). This estimator relies on the assumption that, within sufficiently small neighborhoods, the distribution of distances between nearby points reflects the local dimensionality of the underlying manifold. By examining how distances grow within these local neighborhoods, the estimator provides a statistically grounded estimate of intrinsic dimension.

Formally, let $T_j(x_i)$ denote the distance from a sample x_i to its $j - th$ nearest neighbor. For a fixed neighborhood size k, the local intrinsic-dimension estimate at point x_i is defined as:

$$m(x_i) = \left[\frac{1}{k-1}\sum_{j=1}^{k-1}\log\left(\frac{T_k(x_i)}{T_j(x_i)}\right)\right]^{-1}$$

The overall intrinsic dimension of the dataset is then obtained by averaging the local estimates across all observations:

$$\widehat{m} = \frac{1}{n}\sum_{i=1}^{n} m(x_i)$$

Prior to applying the kNN estimator, the same preprocessing pipeline used throughout the project was applied to the data. Constant features were removed, and the remaining features were standardized to ensure comparable distance scales across dimensions. No additional normalization procedures were applied.

Because the MNIST dataset contains a large number of samples, the kNN estimator was computed on a random subset of 1000 observations. This sampling step reduces computational complexity while maintaining a representative subset of the data for intrinsic-dimension estimation.

A primary estimate was computed using a neighborhood size of k = 10, which is commonly used in practice as a balance between bias and variance in the estimator. In addition, a sensitivity analysis was performed over the range $k \in \{5,\ 8,\ 10,\ 12,\ 15,\ 20\}$, allowing the stability of the estimator to be evaluated with respect to the choice of neighborhood size.

Together with PCA-based dimensionality estimation and correlation-dimension analysis, the kNN-based estimator provides a complementary local geometric perspective on dataset complexity. The combination of these methods enables a more comprehensive assessment of intrinsic dimensionality across datasets with different structural properties.

4.Results and Figure Analysis

This section presents the empirical results obtained from the intrinsic-dimension analysis conducted on the four datasets considered in this study. For each dataset, the results are analyzed using three complementary approaches: PCA-based dimensionality analysis, correlation-dimension estimation, and a k-nearest neighbors–based intrinsic-dimension estimator.

The figures associated with each dataset illustrate different aspects of the data structure, including variance distribution, geometric scaling behavior, and the stability of the intrinsic-dimension estimates.

4.1.2D Plane Embedded in 10D

This dataset represents a simple linear manifold with a known intrinsic dimensionality of two, embedded in a ten-dimensional ambient space. It therefore serves as a useful validation case for evaluating whether the intrinsic-dimension estimators can correctly recover the underlying structure of the data.

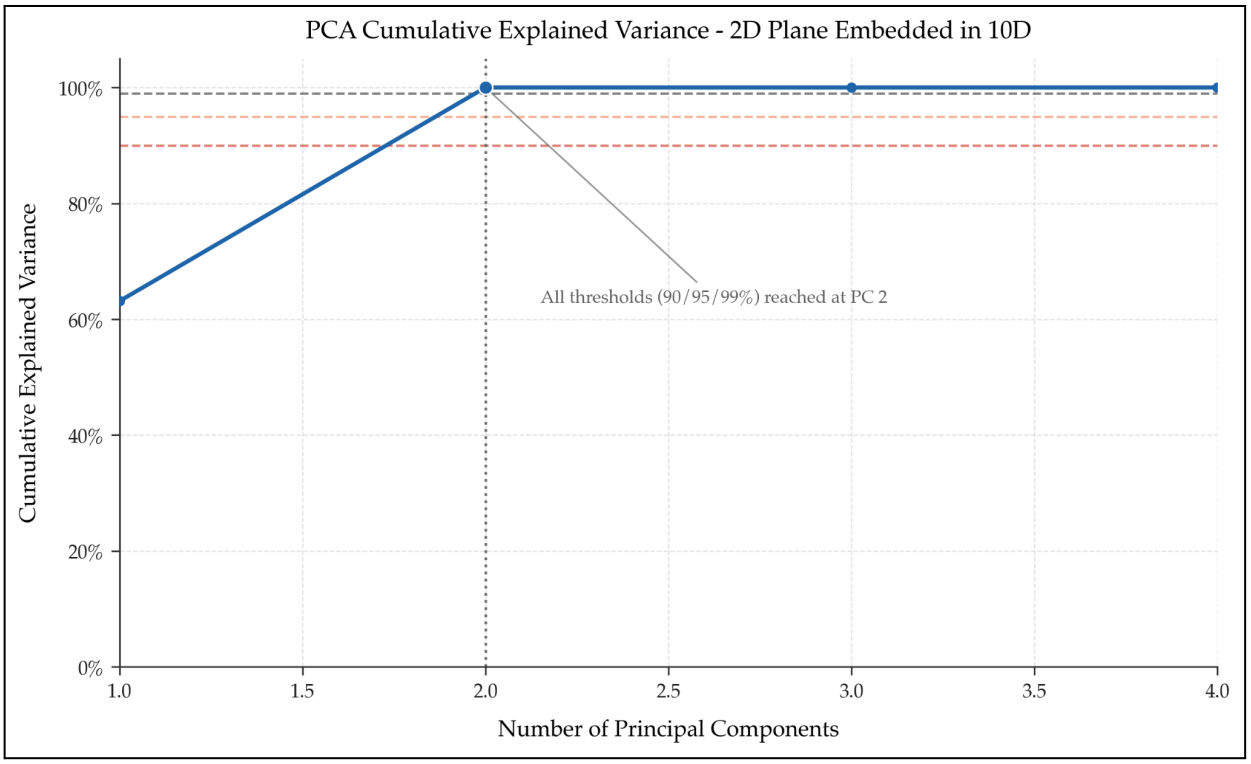

Figure 1. *Cumulative explained variance curve for the 2D Plane Embedded in 10D dataset.*

The cumulative explained variance curve shows that nearly all of the total variance in the dataset is captured by the first two principal components. This behavior is consistent with the expected structure of a two-dimensional linear manifold embedded within a higher-dimensional ambient space.

The rapid concentration of variance in the first two components indicates that the data effectively lie in a low-dimensional linear subspace. After the second component, the marginal contribution of additional principal components becomes negligible, suggesting that the remaining dimensions primarily capture noise introduced by the embedding rather than meaningful variability in the data.

This result therefore confirms that PCA successfully identifies the underlying linear structure of the dataset and accurately recovers its intrinsic dimensionality.

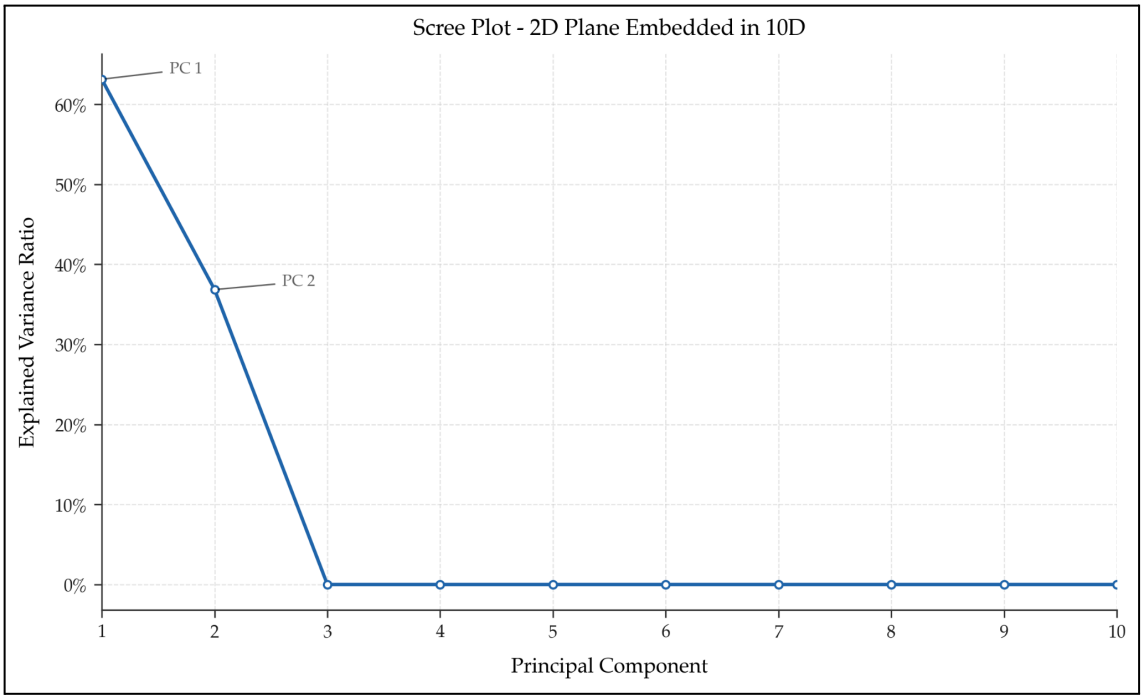

Figure 2. *Scree plot for the 2D Plane Embedded in 10D dataset.*

The scree plot reveals a clear dominance of the first two principal components, while the remaining components contribute only negligible amounts of variance. This sharp decline in eigenvalues after the second component indicates that the majority of the dataset's variability is concentrated in a two-dimensional subspace.

Such a pattern is characteristic of data that lie on a low-dimensional linear manifold embedded within a higher-dimensional ambient space. The scree plot therefore provides strong visual evidence that the intrinsic dimensionality of the dataset is approximately two.

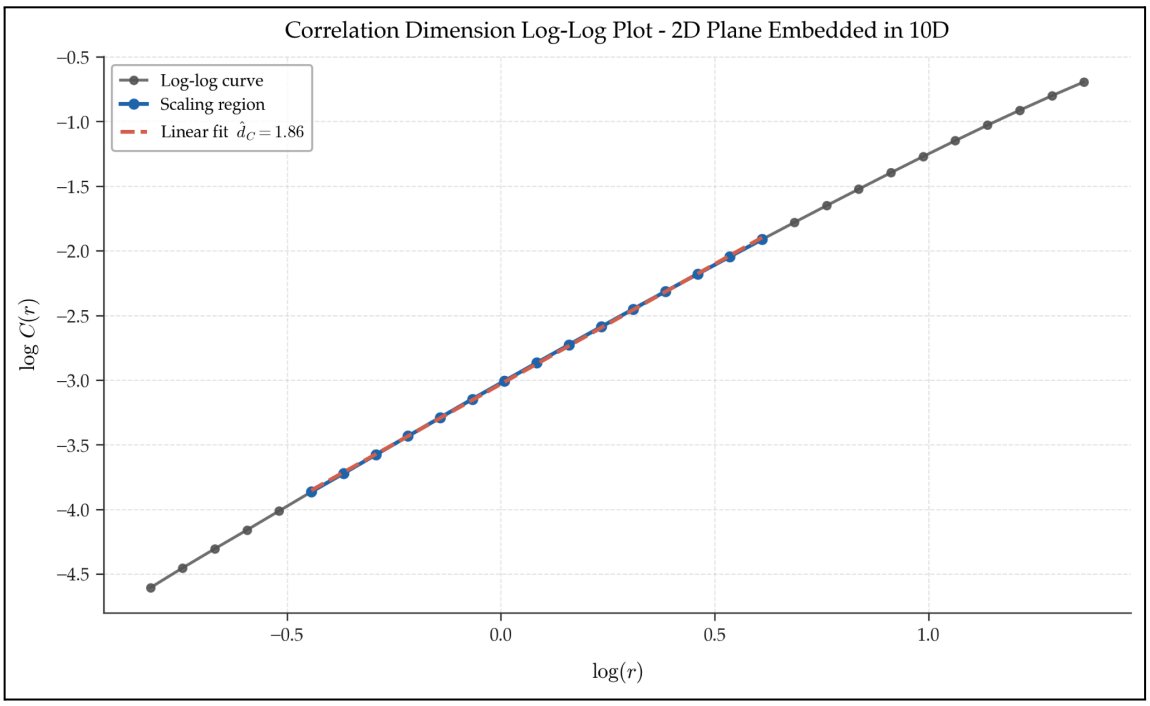

Figure 3. *Correlation-dimension log-log plot for the 2D Plane Embedded in 10D dataset.*

The log–log plot of the correlation integral exhibits a clear linear scaling region, from which the correlation-dimension estimate is obtained. The resulting estimate is very close to two, which is consistent with the theoretical intrinsic dimensionality of the dataset.

This result confirms that the geometric scaling behavior of the data corresponds to that of a two-dimensional manifold embedded in a higher-dimensional space. Moreover, the close agreement between the correlation-dimension estimate

and the PCA-based analysis provides strong validation for both the dataset construction and the overall analysis pipeline.

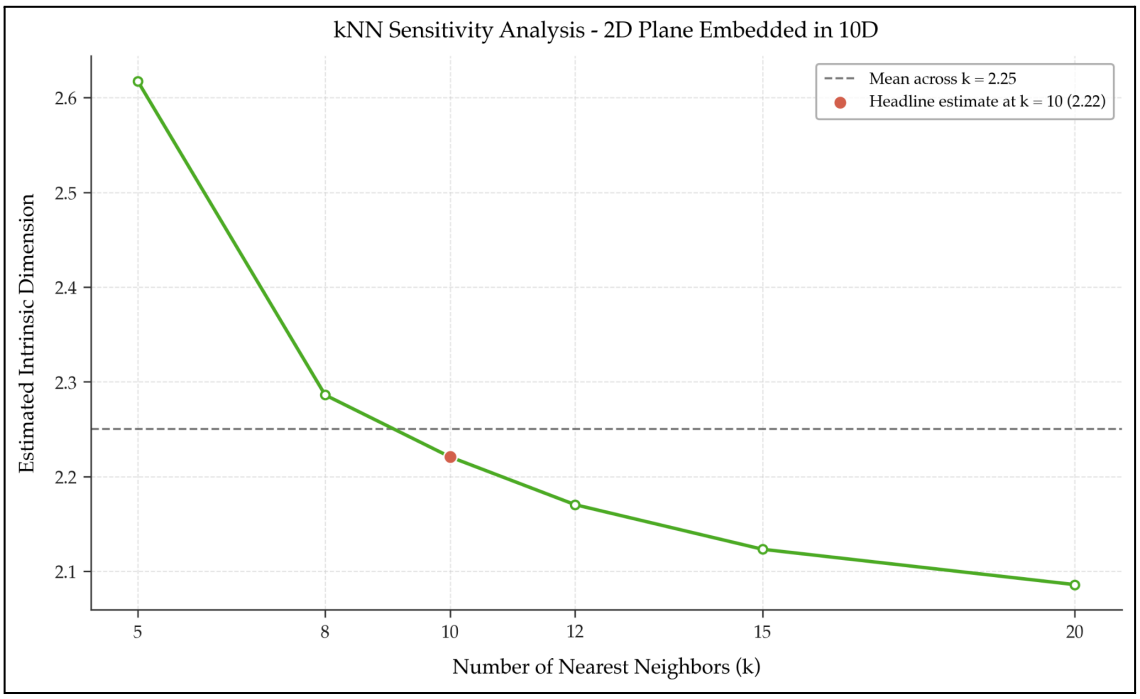

Figure 4. *kNN sensitivity analysis for the 2D Plane Embedded in 10D dataset across multiple values of k.*

The kNN sensitivity analysis shows that the estimated intrinsic dimension remains consistently close to two across the full range of tested neighborhood sizes. Although minor fluctuations appear for smaller values of k, the estimates rapidly stabilize as k increases.

This stability indicates that the Levina–Bickel estimator provides a robust estimate of intrinsic dimensionality for this dataset. The results therefore further support the conclusion that the data lie on a two-dimensional linear manifold embedded in a higher-dimensional ambient space.

4.2.Swiss Roll

This subsection presents the visual analysis results obtained for the Swiss Roll dataset. The Swiss Roll is a classical example of a nonlinear manifold: although the data lie on a two-dimensional surface, the manifold is geometrically curved and embedded in a higher-dimensional space. As a result, this dataset provides an important test case for evaluating the differences between linear and geometric intrinsic-dimension estimators.

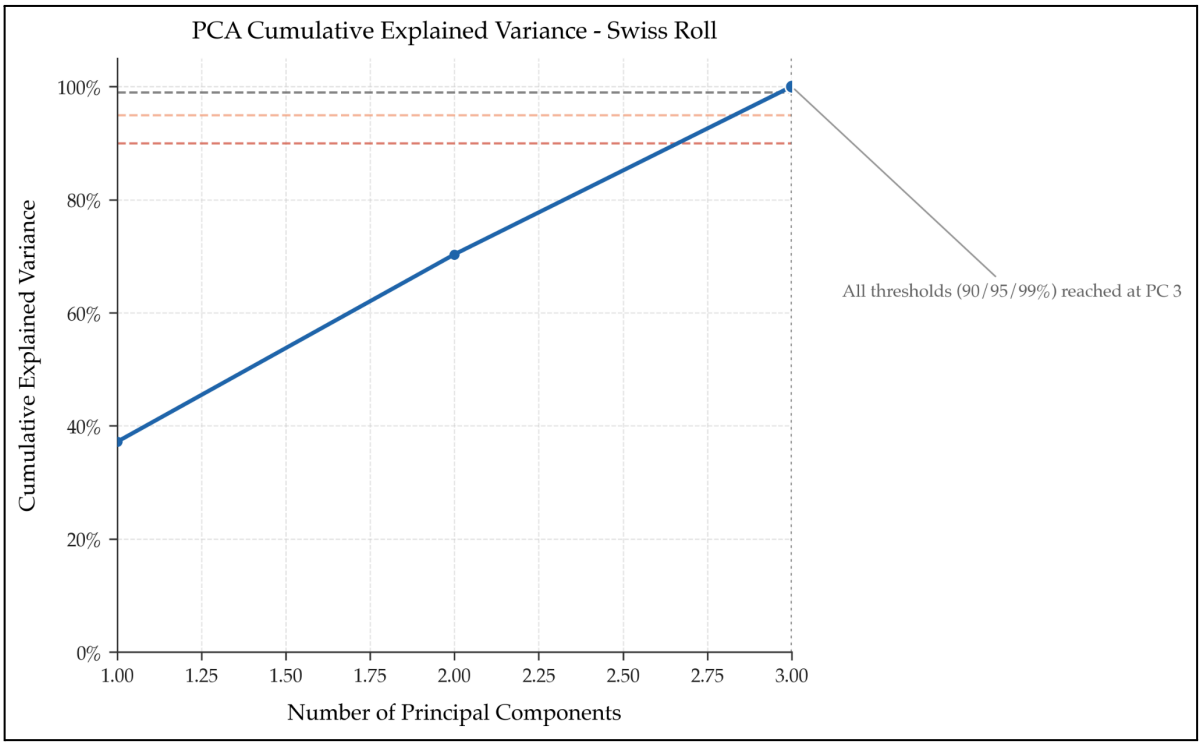

Figure 5. *Cumulative explained variance curve for the Swiss Roll dataset.*

The cumulative explained variance curve indicates that several principal components are required to capture most of the variance in the dataset. Unlike the linear plane dataset analyzed previously, the variance is not concentrated entirely in the first two components.

This behavior occurs because PCA is restricted to linear projections. Since the Swiss Roll manifold is geometrically curved, a larger number of linear components is required to approximate the nonlinear structure of the data. Consequently, PCA tends to overestimate the effective dimensionality of such manifolds.

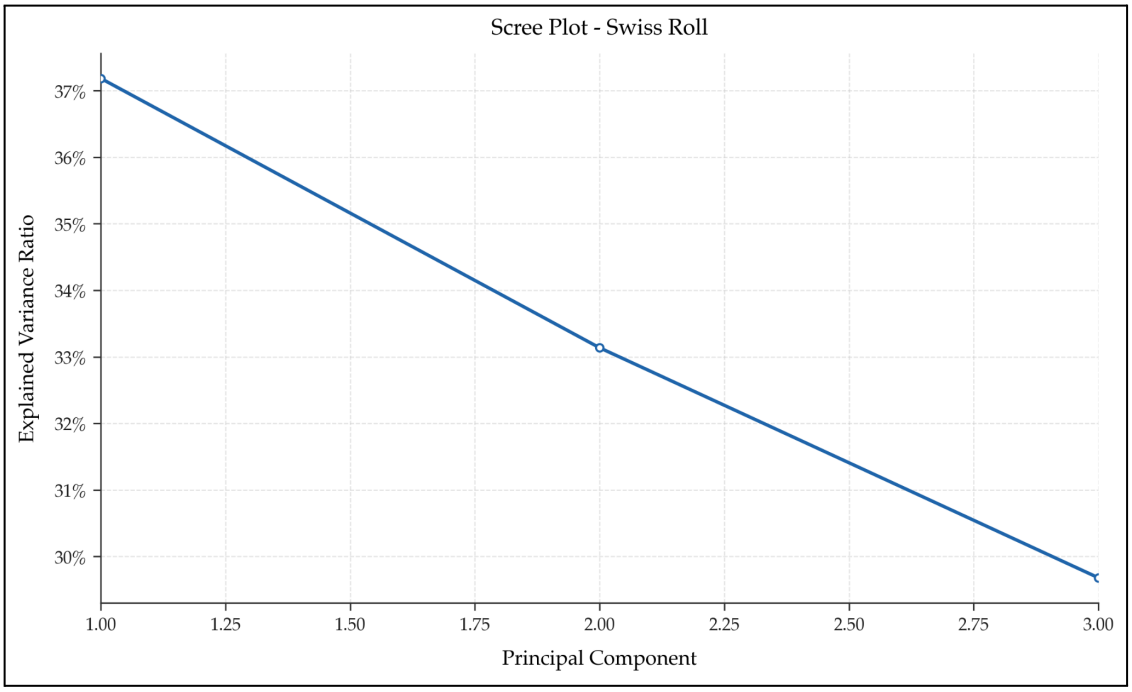

Figure 6. *Scree plot for the Swiss Roll dataset.*

The scree plot shows a gradual decay of eigenvalues rather than a sharp drop after the first two components. This pattern indicates that the variance is distributed across multiple linear directions.

Such behavior is characteristic of nonlinear manifolds when analyzed using PCA. Although the intrinsic structure of the dataset is two-dimensional, the curvature of the manifold causes the variance to be spread across several principal components in the linear embedding space.

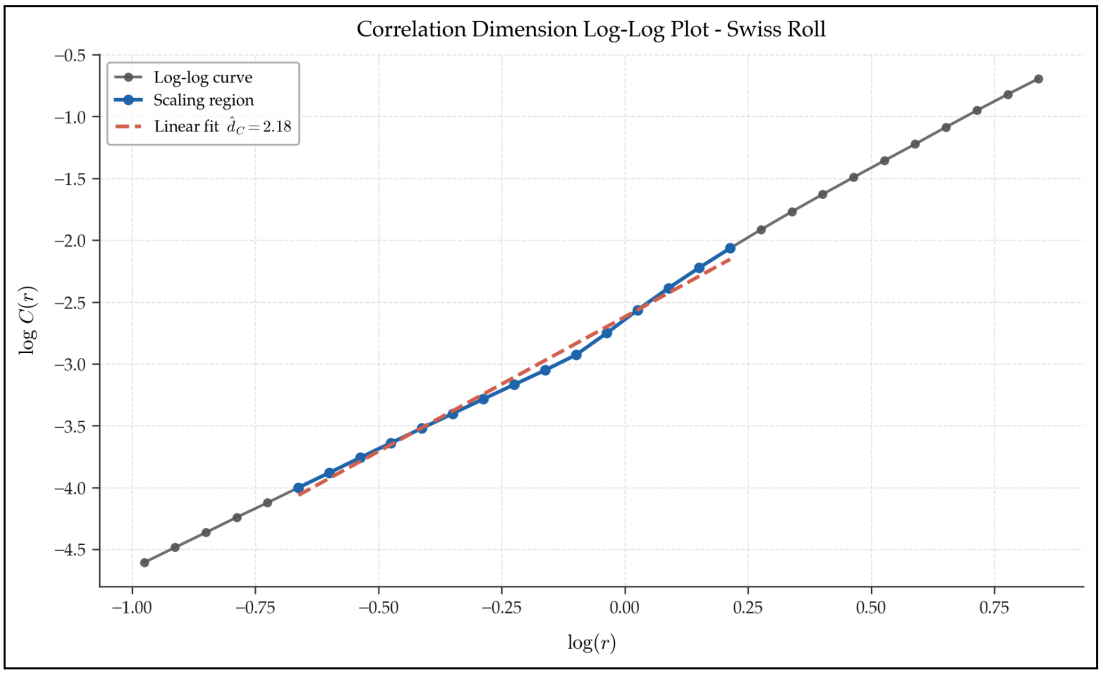

Figure 7. *Correlation-dimension log-log plot for the Swiss Roll dataset.*

The log–log plot of the correlation integral exhibits a clear linear scaling region from which the correlation dimension can be estimated. The resulting estimate is close to two, which aligns with the known intrinsic dimensionality of the Swiss Roll manifold.

This result demonstrates that geometric scaling methods are able to recover the true dimensional structure of nonlinear manifolds, even when linear methods such as PCA fail to do so.

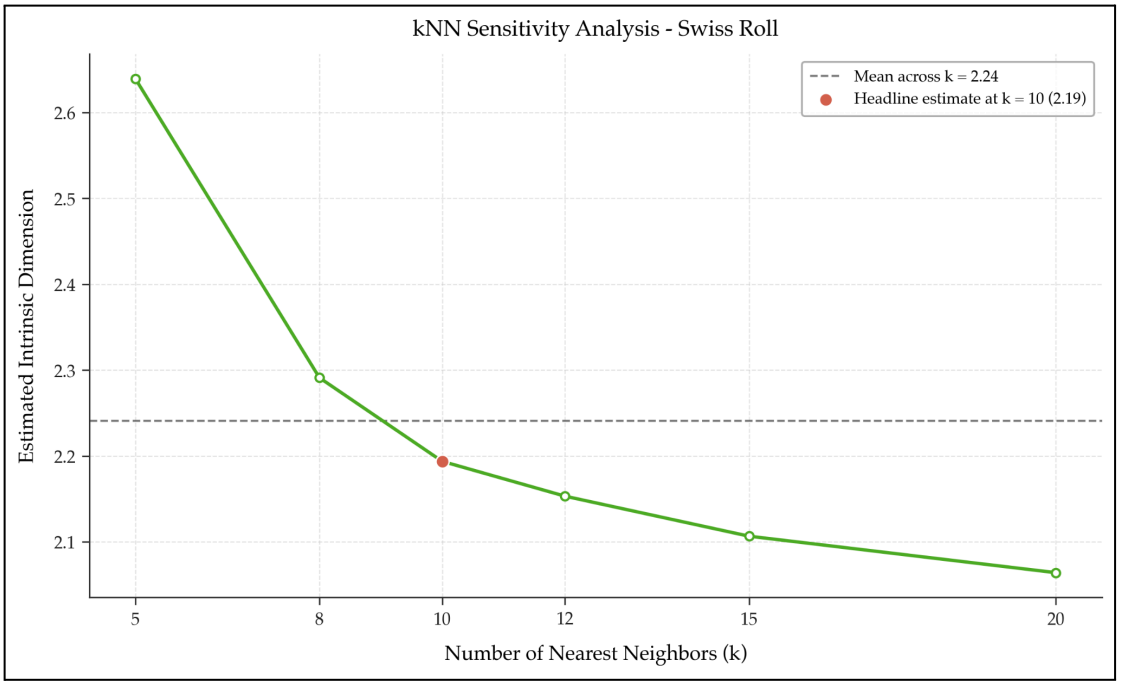

Figure 8. *kNN sensitivity analysis for the Swiss Roll dataset across multiple values of k.*

The kNN-based intrinsic dimension estimates remain consistently close to two across the tested range of neighborhood sizes. The estimator exhibits only minor fluctuations for smaller values of k, after which the estimates stabilize.

Together with the correlation-dimension results, this provides strong evidence that the Swiss Roll dataset is intrinsically two-dimensional despite its nonlinear embedding in a higher-dimensional ambient space.

4.3.Gaussian Noise

This subsection presents the visual analysis results obtained for the Gaussian Noise dataset. In contrast to the structured datasets examined earlier, this dataset represents unstructured high-dimensional data generated from an isotropic Gaussian distribution. As such, it provides an important baseline for distinguishing between datasets that possess an underlying low-dimensional manifold and those that do not.

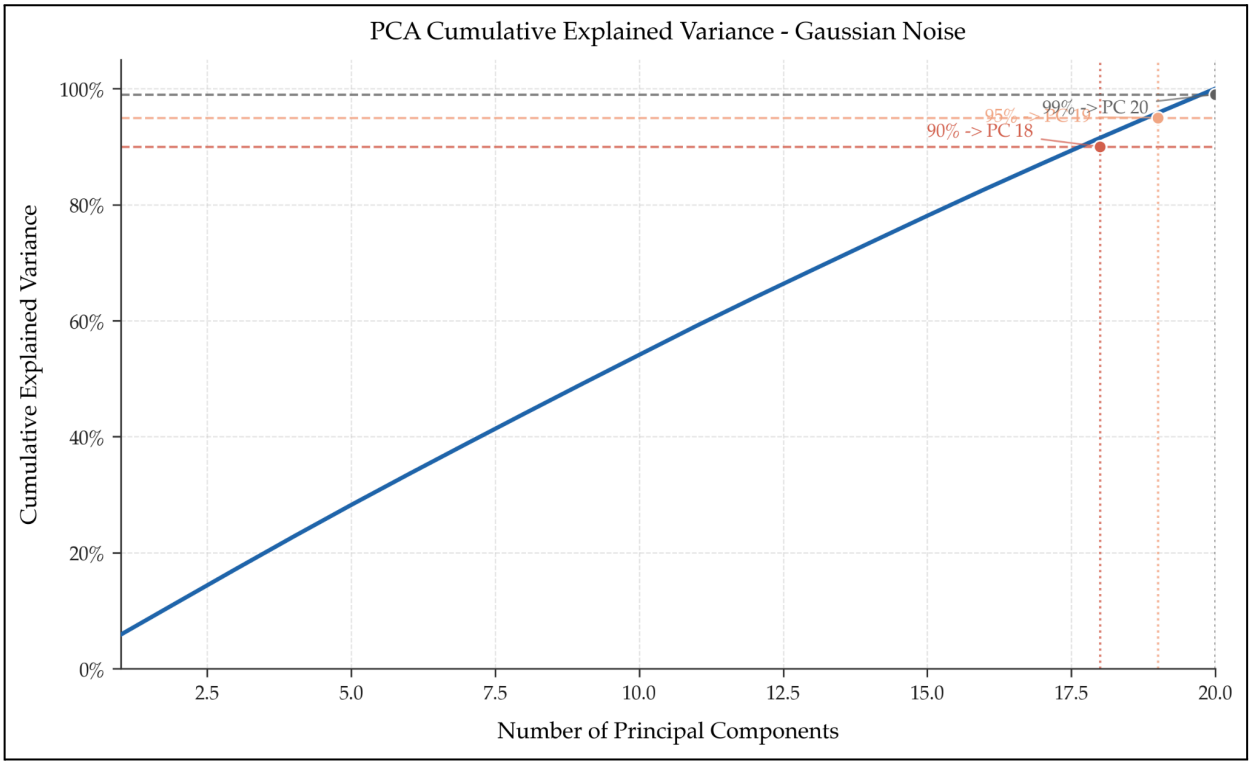

Figure 9. *Cumulative explained variance curve for the Gaussian Noise dataset.*

The cumulative explained variance curve increases gradually rather than sharply. Unlike the previous datasets, no small subset of principal components captures a dominant portion of the variance.

This behavior indicates that the variance is distributed across many directions in the feature space. Such a pattern is characteristic of isotropic high-dimensional noise, where no compact low-dimensional linear subspace governs the structure of the data.

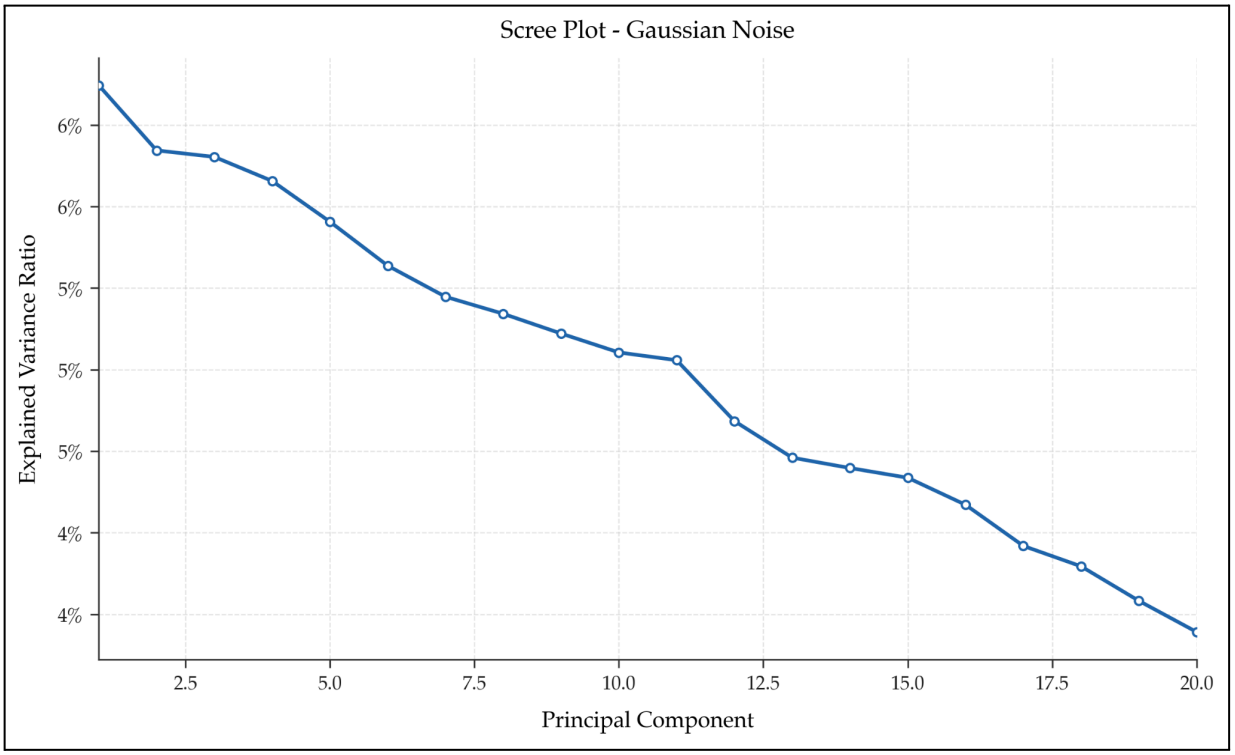

Figure 10. *Scree plot for the Gaussian Noise dataset.*

The scree plot shows a relatively smooth decay of eigenvalues without a clear elbow point. No small number of components dominates the variance structure, and the eigenvalues decrease gradually across the full range of principal components.

This pattern further confirms the absence of a strong low-dimensional linear representation, which is consistent with the expected behavior of randomly generated high-dimensional noise.

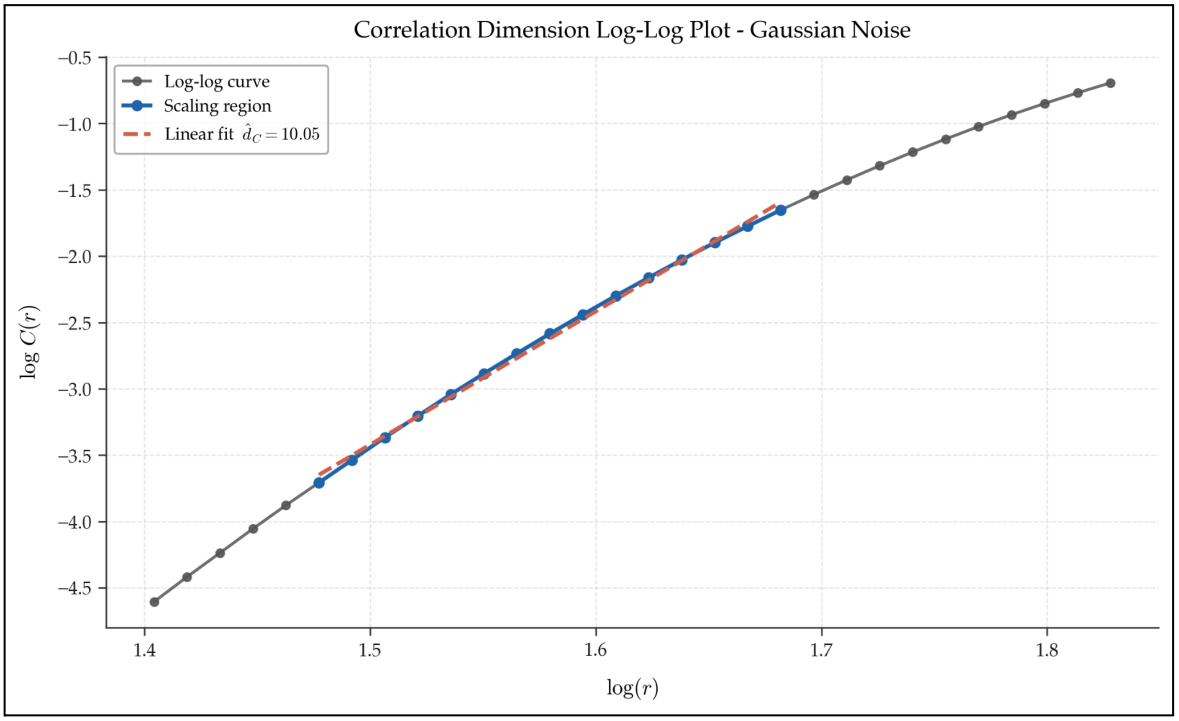

Figure 11. *Correlation-dimension log-log plot for the Gaussian Noise dataset.*

The correlation-dimension analysis yields a comparatively high estimate relative to the other datasets examined in this study. This outcome is consistent with the fact that the data do not lie on a compact low-dimensional manifold.

Instead, the scaling behavior of pairwise distances reflects the diffuse geometric structure of high-dimensional noise. Together with the PCA results, the correlation-dimension analysis clearly distinguishes the Gaussian Noise dataset from

the structured synthetic manifolds and from the MNIST dataset.

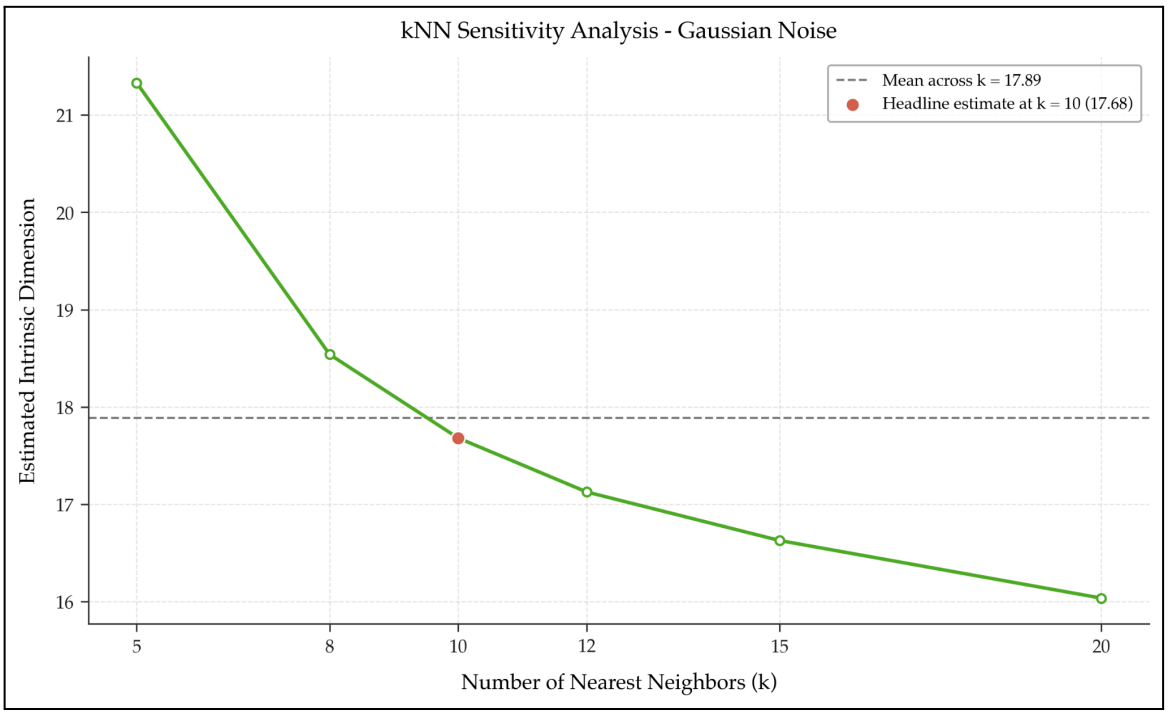

Figure 12. *kNN sensitivity analysis for the Gaussian Noise dataset across multiple values of k.*

The kNN-based intrinsic dimension estimates remain in a relatively high-dimensional regime across all tested neighborhood sizes. In comparison with the low-dimensional synthetic datasets, the estimates exhibit slightly greater variability across values of k.

Such variability is expected when analyzing datasets that do not lie on a sharply defined low-dimensional manifold. Nevertheless, the overall magnitude of the estimates remains consistent with the high intrinsic dimensionality expected for isotropic Gaussian noise.

4.4.MNIST

This subsection presents the visual analysis results obtained for the MNIST dataset. Unlike the synthetic datasets analyzed earlier, MNIST represents a real-world high-dimensional dataset consisting of handwritten digit images. Each observation is represented as a 784-dimensional vector corresponding to the pixel intensities of a 28 \times 28 grayscale image.

Although the ambient dimensionality is very high, the generative process underlying handwritten digits is expected to be governed by a much smaller number of degrees of freedom, such as stroke shape, orientation, thickness, and writing style. Intrinsic dimension estimation therefore provides an important tool for understanding the effective geometric complexity of the dataset.

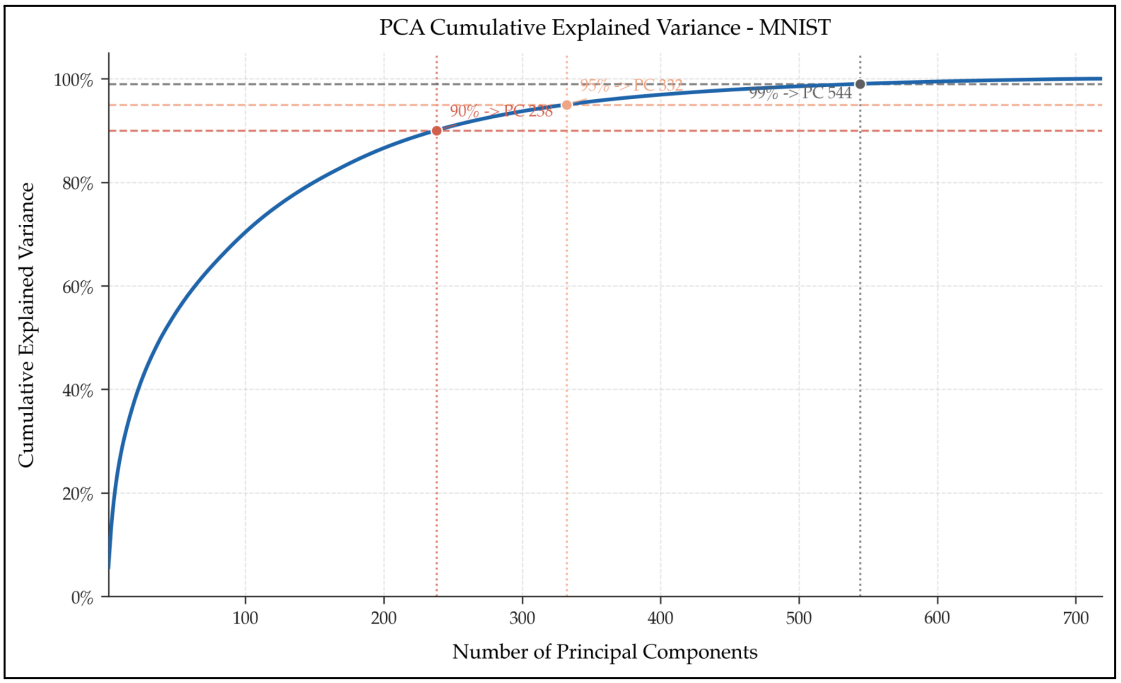

Figure 13. *Cumulative explained variance curve for the MNIST dataset.*

The cumulative explained variance curve shows that a large number of principal components is required to retain a high proportion of the total variance. In particular, hundreds of components are needed to capture 90%, 95%, and 99% of the variance.

This behavior reflects the fact that variability in raw pixel space is distributed across many linear directions. Differences in digit shape, stroke curvature, writing thickness, and spatial positioning all contribute to variance that cannot be captured by only a small number of linear components.

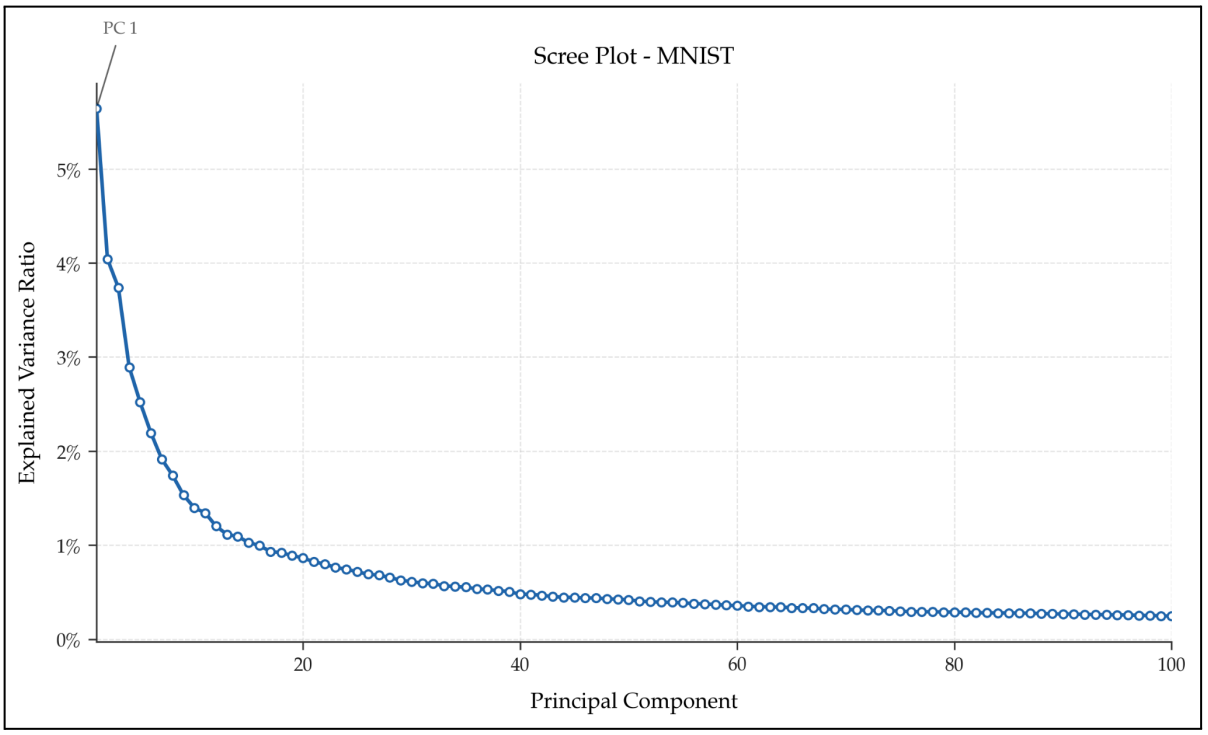

Figure 14. *Scree plot for the MNIST dataset.*

The scree plot shows a gradual decay of eigenvalues rather than a sharp drop after a small number of components. This indicates that many principal components contribute non-negligibly to the overall variance of the dataset.

Such behavior is typical for complex image datasets, where variability arises from multiple interacting factors. Although MNIST digits share a common semantic structure, the diversity of handwriting styles leads to a relatively diffuse distribution of variance across principal components.

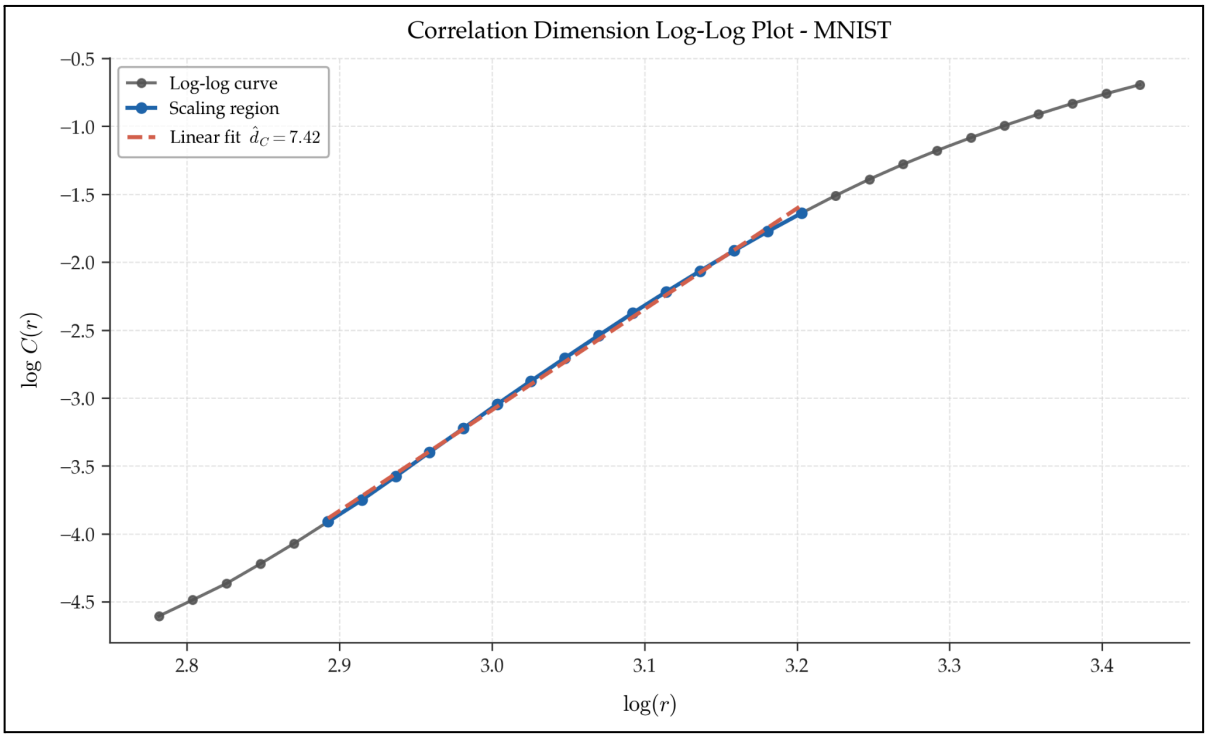

Figure 15. *Correlation-dimension log-log plot for the MNIST dataset.*

The correlation-dimension analysis yields an estimate that is substantially lower than the number of principal components required to capture most of the variance. This difference highlights the distinction between linear variance structure and intrinsic geometric dimensionality.

While PCA measures how variance is distributed across linear directions, the correlation dimension reflects the local scaling behavior of pairwise distances. The lower estimate therefore suggests that MNIST occupies a structured

nonlinear region within the high-dimensional pixel space rather than filling the ambient space uniformly.

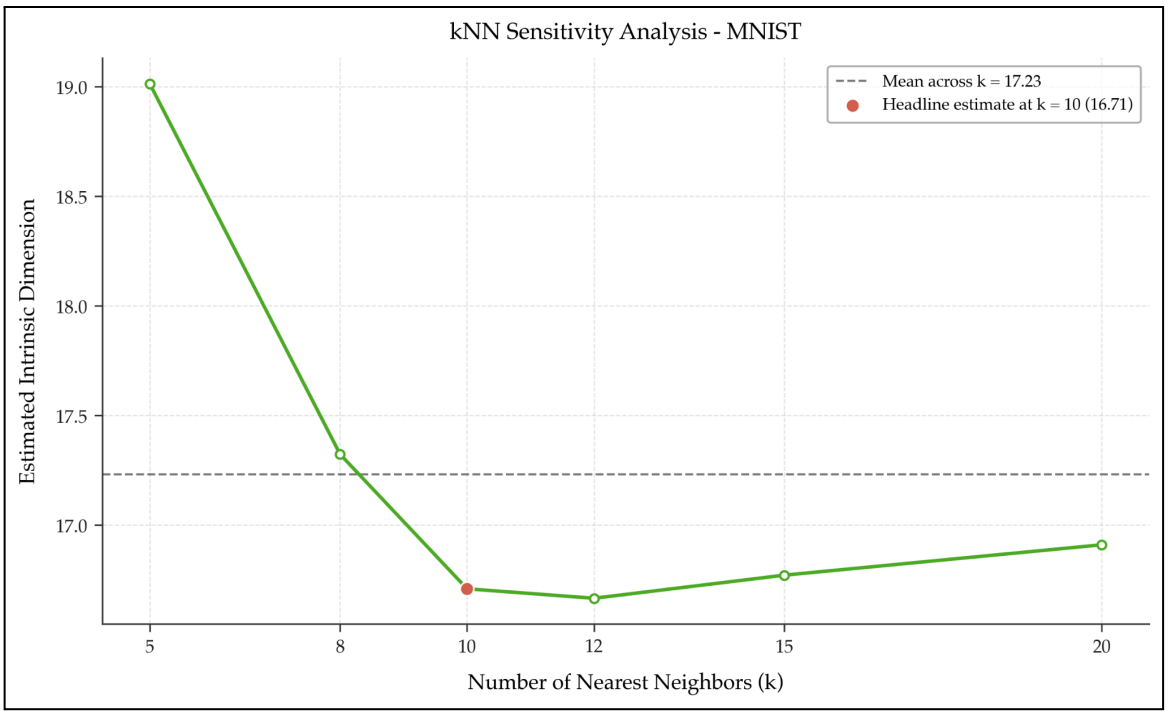

Figure 16. *kNN sensitivity analysis for the MNIST dataset across multiple values of k.*

The kNN-based intrinsic dimension estimates fall in the high-teen range and remain relatively stable across the tested neighborhood sizes. This stability indicates that the Levina–Bickel estimator provides a consistent estimate of the dataset's intrinsic dimensionality.

Notably, the kNN estimate lies between the correlation-dimension estimate and the PCA-based dimensionality estimate. This intermediate result supports the interpretation that MNIST possesses a structured yet moderately complex intrinsic geometry: the data do not occupy the full 784-dimensional ambient space, but they also cannot be represented by only a few linear components.

5.Global Comparison Across Datasets

After examining each dataset individually, it is useful to compare the dimensionality estimates obtained from the three methods applied in this study: PCA-based dimensionality estimation, correlation dimension, and the k-nearest neighbors (kNN) estimator. Each method captures a different aspect of the geometric structure of the data, and a global comparison helps clarify how these perspectives relate across datasets with different underlying properties.

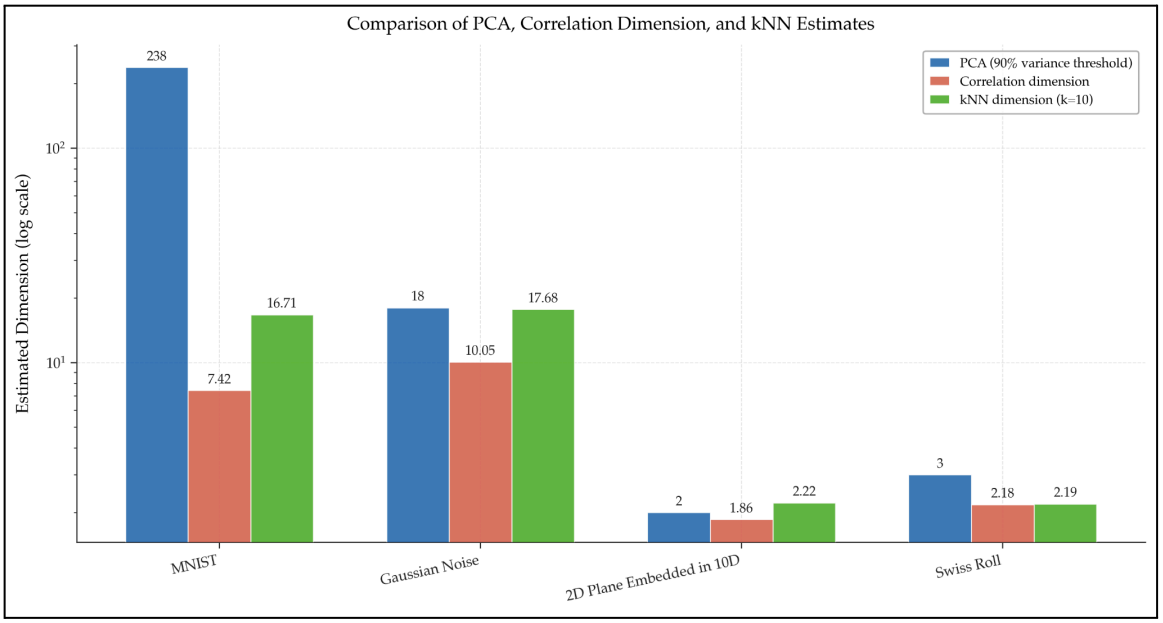

Figure 17. *Comparison of PCA, correlation-dimension, and kNN-based dimensionality estimates across all datasets.*

The full comparison highlights the complementary nature of the three estimation approaches. PCA focuses on the distribution of variance across linear directions, whereas the correlation dimension characterizes the local geometric scaling of the dataset. The kNN-based estimator provides an intermediate perspective by estimating intrinsic dimension from the statistics of nearest-neighbor distances.

Across the datasets analyzed in this project, these methods reveal consistent structural patterns. PCA generally produces higher dimensionality estimates because it approximates the dataset using linear components, while the correlation dimension often yields lower values reflecting the underlying geometric manifold. The kNN estimator typically produces estimates that lie between these two perspectives.

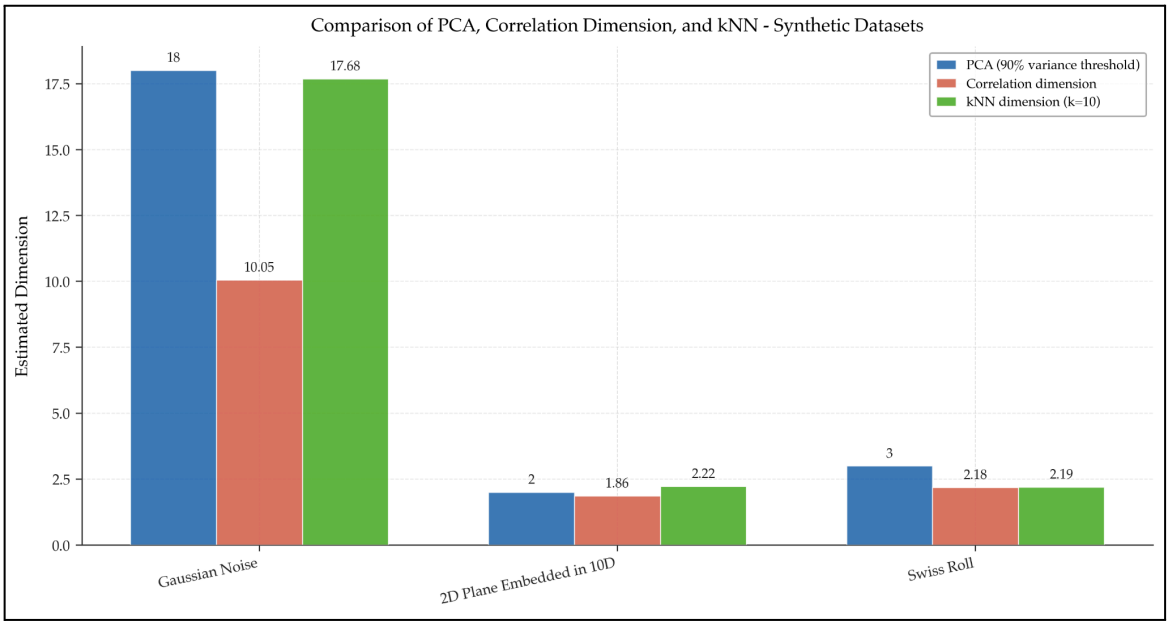

Figure 18. *Comparison of PCA, correlation-dimension, and kNN estimates restricted to the synthetic datasets.*

Restricting the comparison to the synthetic datasets provides a clearer view of how the estimators behave when the underlying geometric structure is known. In particular, both the two-dimensional plane embedded in 10D and the Swiss Roll dataset exhibit intrinsic dimensionality close to two according to the geometric estimators.

In contrast, the Gaussian Noise dataset remains high-dimensional across all estimation methods. This comparison illustrates how intrinsic-dimension estimators can effectively distinguish between structured manifolds and unstructured high-dimensional noise.

5.1.kNN Summary Table

The following table summarizes the kNN-based intrinsic dimension estimates obtained for each dataset.

Dataset	Samples Used	Processed Dimension	kNN Estimate (k=10)	Mean Across k	Std. Across k
MNIST (sampled)	1000	599	16.71	17.23	0.83
Gaussian Noise	2000	20	17.68	17.89	1.73
2D Plane Embedded in 10D	2000	10	2.22	2.25	0.18
Swiss Roll	2000	3	2.19	2.24	0.19

The kNN-based results are highly consistent with the expected structure of the datasets. For the two-dimensional plane embedded in 10D and the Swiss Roll dataset, the kNN estimates are both very close to two, confirming that these datasets lie on intrinsically two-dimensional manifolds despite their embedding in higher-dimensional ambient spaces.

For the Gaussian Noise dataset, the kNN estimate remains relatively high, which is consistent with the absence of a compact low-dimensional manifold. In the case of MNIST, the estimated intrinsic dimension is substantially lower than the ambient dimensionality of the pixel space, yet significantly higher than the intrinsic dimensionality of the synthetic manifolds. This result indicates that MNIST exhibits a structured but moderately complex intrinsic geometry.

5.2.Visual Illustration of the MNIST Dataset

To provide visual context for the type of data analyzed in this study, a sample of MNIST digit images is shown below.

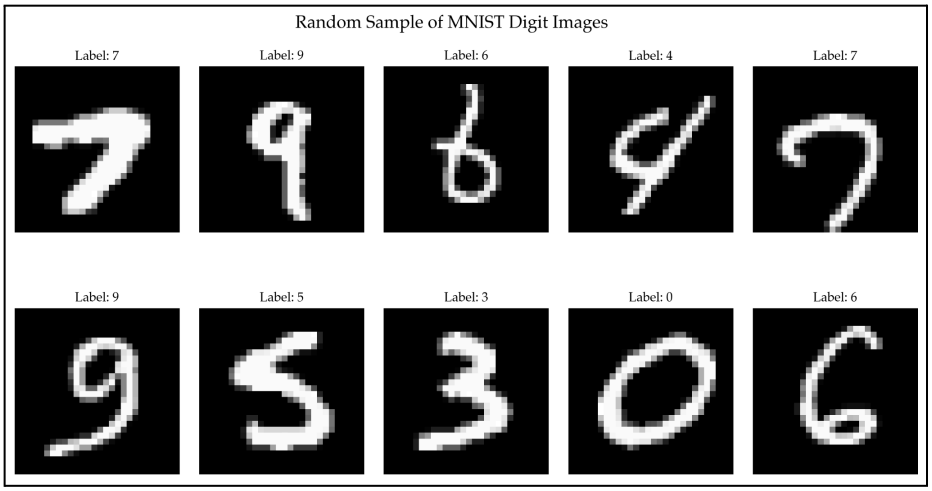

Figure 19. *Random sample of MNIST handwritten digit images.*

Although each image is represented numerically as a 784-dimensional vector in pixel space, the visual variation among the images is governed by a much smaller set of semantic factors. These factors include digit identity, stroke thickness, curvature, orientation, and individual writing style.

This observation supports the hypothesis that the MNIST dataset occupies a structured low-dimensional manifold within

the high-dimensional ambient space defined by the pixel representation.

6.Discussion and Conclusion

The results obtained in this study demonstrate that intrinsic dimensionality cannot be characterized by a single universal estimator. Instead, different estimation methods capture different aspects of the structure of high-dimensional datasets. The comparison between PCA-based dimensionality estimation, correlation dimension, and k-nearest neighbors (kNN) estimators highlights the importance of combining multiple analytical perspectives when studying the geometry of complex data.

From a methodological standpoint, the three approaches examined in this project differ fundamentally in what they measure. PCA is a variance-based linear method that identifies directions explaining the largest proportion of variance in the dataset. As a result, it is most informative when the data lie close to a linear subspace. In contrast, the correlation dimension is a geometric measure that characterizes how the number of neighboring points scales as the observation radius decreases. This makes it particularly well suited for capturing the intrinsic geometry of nonlinear manifolds. The kNN-based estimator provides a complementary perspective by estimating intrinsic dimension directly from local nearest-neighbor distances, offering a neighborhood-based approximation of the dataset's geometric complexity.

The behavior of the estimators across the datasets analyzed in this project illustrates these methodological differences clearly. For the two-dimensional plane embedded in a ten-dimensional space, all methods consistently recovered an intrinsic dimensionality close to two. This dataset therefore served as an effective sanity check for the implementation and validation of the analysis pipeline.

The Swiss Roll dataset provided a more challenging example involving nonlinear geometry. In this case, PCA required multiple components to approximate the variance structure, while the correlation-dimension and kNN estimators correctly identified the underlying two-dimensional manifold. This result highlights the well-known limitation of linear dimensionality methods when applied to curved manifolds.

The Gaussian Noise dataset behaved fundamentally differently from the structured datasets. Because the data were generated from an isotropic high-dimensional distribution without an underlying manifold, the dimensionality estimates remained relatively high across all methods. This behavior confirms that intrinsic dimension estimators can effectively distinguish between structured manifolds and unstructured high-dimensional noise.

The MNIST dataset provided the most interesting case study in the project. Although the ambient dimensionality of the dataset is very high (784 dimensions), the geometric estimators revealed a substantially lower intrinsic dimensionality. At the same time, the intrinsic dimension estimates for MNIST remained significantly larger than those observed for the simple synthetic manifolds. This suggests that MNIST occupies a structured yet moderately complex region of the high-dimensional pixel space, reflecting the diversity of handwriting styles and the nonlinear factors governing digit formation.

These observations are closely related to the **Manifold Hypothesis,** which states that many high-dimensional real-world datasets lie near low-dimensional manifolds embedded in high-dimensional ambient spaces. The results obtained for MNIST are consistent with this hypothesis: although the data are represented in a very high-dimensional space, their effective geometric structure appears to be governed by a much smaller number of degrees of freedom.

Overall, the combination of PCA-based analysis, correlation-dimension estimation, and kNN-based intrinsic dimension estimation provides a more complete and balanced understanding of dataset structure than any single method alone. By integrating linear variance analysis with geometric scaling methods and local neighborhood statistics, it becomes possible to obtain deeper insight into the underlying complexity of high-dimensional data.

This project therefore demonstrates how intrinsic-dimension estimation can serve as a valuable tool for understanding the geometric structure of both synthetic and real-world datasets.